

Stick Lengths

Time limit: 1.00 s

Memory limit: 512 MB

There are n sticks with some lengths. Your task is to modify the sticks so that each stick has the same length.

You can either lengthen and shorten each stick. Both operations cost x where x is the difference between the new and original length.

What is the minimum total cost?

Input

The first input line contains an integer n : the number of sticks.

Then there are n integers: $p_1, p_2, \dots, p_n$: the lengths of the sticks.

Output

Print one integer: the minimum total cost.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq p_i \leq 10^9$

Example

Input	Output
5 2 3 1 5 2	5